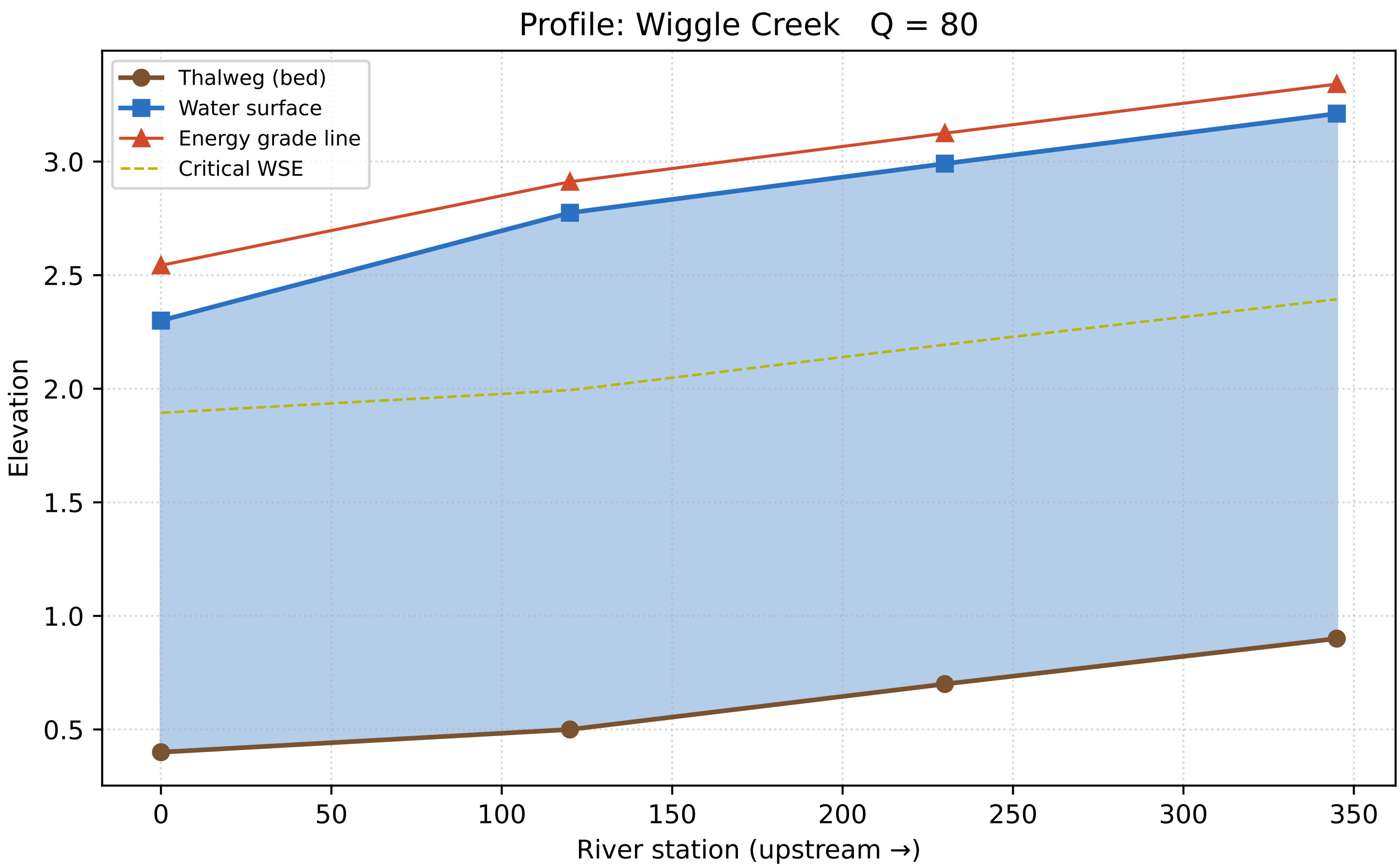
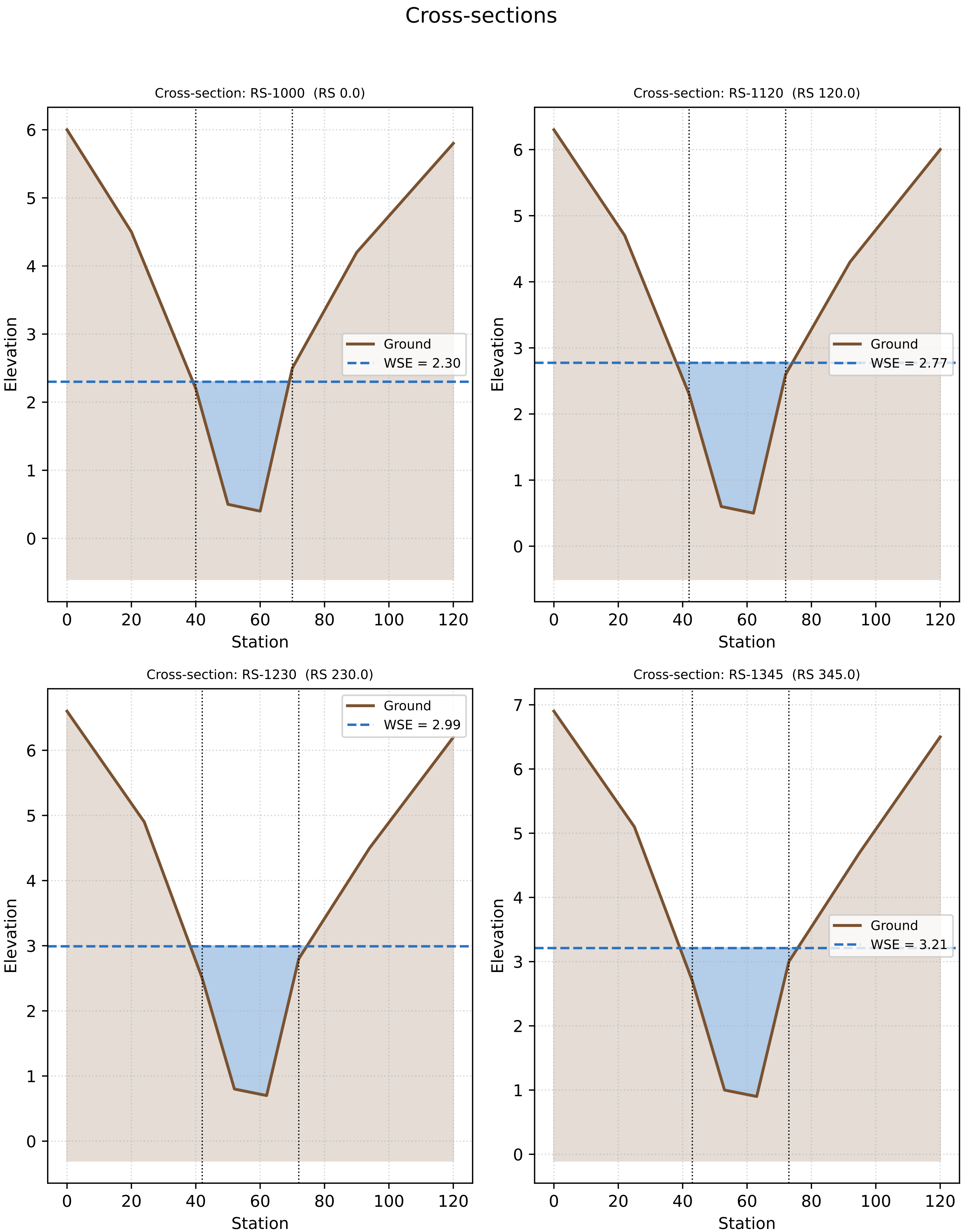
Water-Surface Profile Report

Sample University · CIVE 462 · Reach: Wiggle Creek · Q = 80 · Units: SI · 2026-06-13 13:45
Project: Natural Irregular River



Section	Sta.	WSE	Depth	Vel.	Fr	Energy	Regime	Conv?
RS-1000	0	2.30	1.90	2.18	0.63	2.54	subcritical	yes
RS-1120	120	2.77	2.27	1.64	0.45	2.91	subcritical	yes
RS-1230	230	2.99	2.29	1.62	0.44	3.12	subcritical	yes
RS-1345	345	3.21	2.31	1.60	0.44	3.34	subcritical	yes


Cross-sections
Cross-section: RS-1000 (RS 0.0)
Ground
WSE = 2.30
Elevation
Station
0
20
40
60
80
100
120
0
1
2
3
4
5
6
Cross-section: RS-1120 (RS 120.0)
Ground
WSE = 2.77
Elevation
Station
Cross-section: RS-1230 (RS 230.0)
Ground
WSE = 2.99
Elevation
Station
Cross-section: RS-1345 (RS 345.0)
Ground
WSE = 3.21
Elevation
Station
7

Methodology & Disclaimer

Water-surface elevations were computed with the standard-step method, iterating the 1-D energy equation between adjacent cross-sections. Friction slope is from Manning's equation; conveyance is subdivided by left-overbank / channel / right-overbank panels where bank stations are defined. Critical depth is found by locating the minimum-specific-energy depth for the given discharge. See docs/automation.md and the inline ### LEARN ### notes in mirrorz/hydraulics.py and mirrorz/solver.py.

Disclaimer

This report was produced by MirrorZ-Hecras, an EDUCATIONAL tool for learning open-channel hydraulics. It is NOT certified or validated for regulatory floodplain mapping, or for the design of levees, dams, bridges, culverts, or any structure whose failure could endanger life or property. Results may differ from validated software including HEC-RAS. Engineering decisions must be made by a qualified, licensed professional using validated tools.